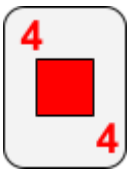
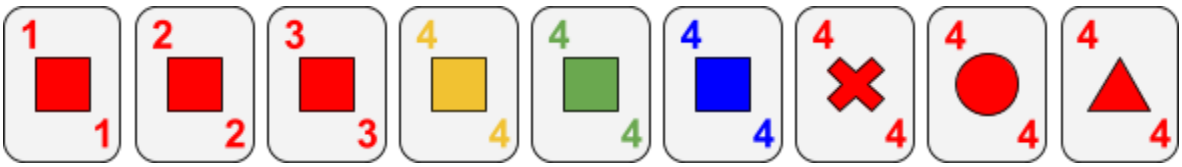
American Computer Science League

2024-2025 ● Contest 2: Uno Duo ● Senior Division

PROBLEM: The card game *Duo* uses cards that have 3 characteristics: a color, number, and symbol. The game starts with two face-up cards and 7 cards in the player's hand. On a single turn, the player can choose one of the two face-up cards and *match* as many cards from their hand as possible, thus making a pile. A card can be placed on top of another card if exactly 2 of the characteristics match.



The ACSL cards use four colors (R=red, Y=yellow, G=green, B=blue), four numbers (1, 2, 3, 4), and four symbols (X=cross, O=circle, S=square, T=triangle). A card is represented by a 3-character string showing the color, number, and symbol, in that order. For example, the card R4S, shown on the left, is red with the number 4 and contains a square. There are 9 cards that could possibly be placed on top of this card: a red square with the number 1, 2, or 3; a yellow, green, or blue square with the number 4; a red cross, circle, or triangle with the number 4.



The deck of cards may also have "wild" cards that have just 1 of these 3 characteristics and would be represented as a string such as a color (e.g. G) or a number (e.g. 3), or a shape (e.g. S). These cards can be matched if that one characteristic matches the other card's same characteristic, regardless of what the other two characteristics are.

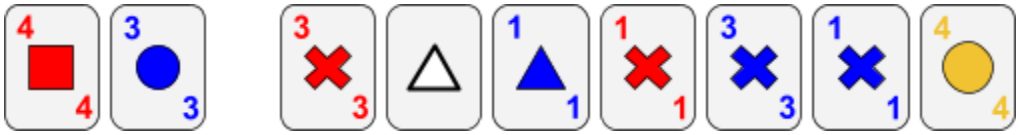
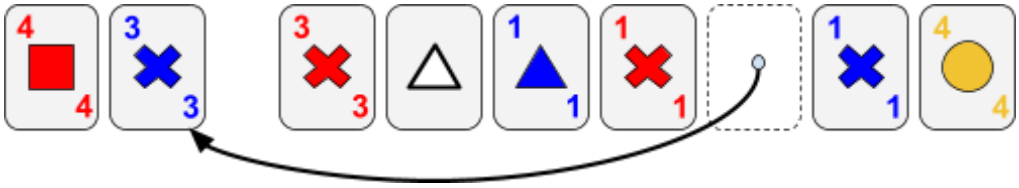
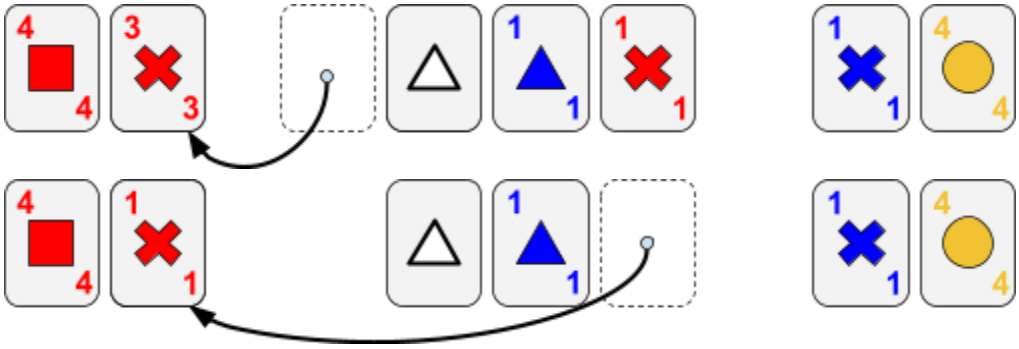
Your program will play multiple turns of Uno Duo for a single player. The turn begins with two face-up cards, 7 cards in the hand (think of them as being held from left to right), and a draw pile.

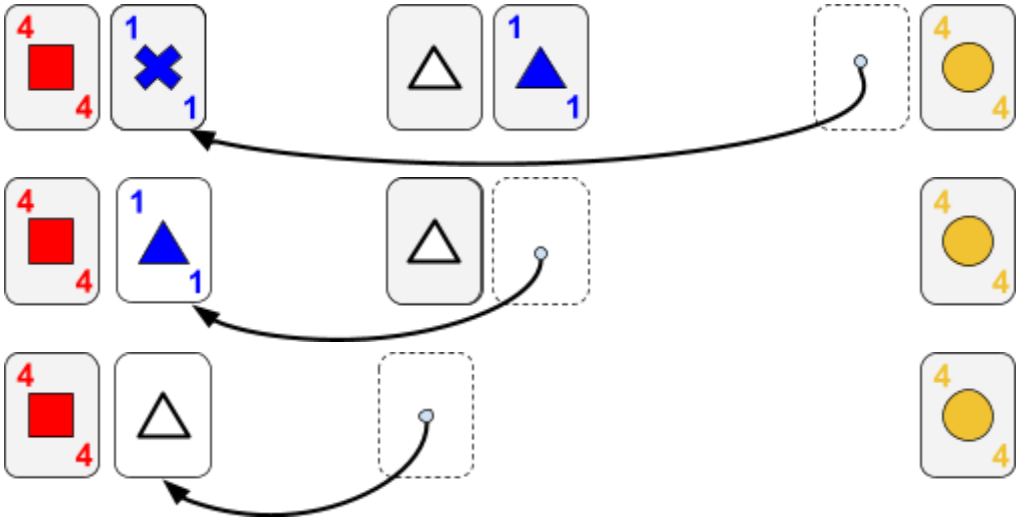
1. Find the leftmost card in your hand that matches one of the face-up cards. Place the matched card on that face-up card, creating a pile. If the card matches both face-up cards, choose the first card.
2. Try to match as many cards as you can in your hand on that pile. Each time, look at your cards starting at the leftmost card.
3. If a Wild Card is at the top of the pile, play the first card that matches it.
4. Your turn is over when there are no more cards in your hand or you cannot play any of them.

When your turn is over, add as many cards as possible from the draw pile so that you have no more than 7 cards in your hand. Continue with another turn until you've played all the cards in your hand and the draw pile is empty, or no cards can be matched. Output the number of cards in your hand, followed by the top card in each pile. Separate each by a single space.

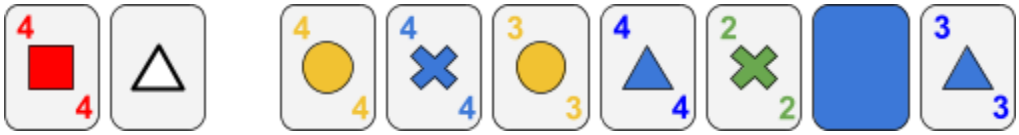
EXAMPLE:

Input	`R4S B3O`
	`R3X T B1T R1X B3X B1X Y4O`
	`B4X Y3O B4T G2X B B3T`

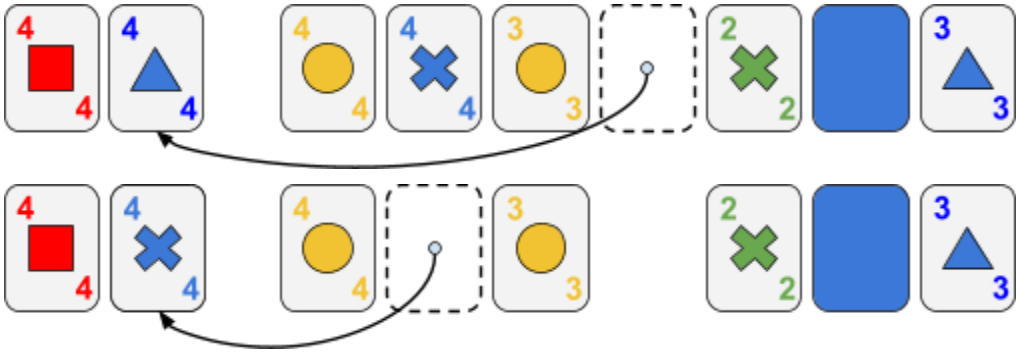
Output	`3 R4S B3T`
Explanation	Initially, the two face-up cards and the 7 cards in the hand are as follows:  The first card in the hand that can be played is B3X (matching the color and the number, but not the shape) on the 2nd face-up card, thus making a pile. All future matches in this turn must go on the 2nd pile.  The next moves are as follows: 

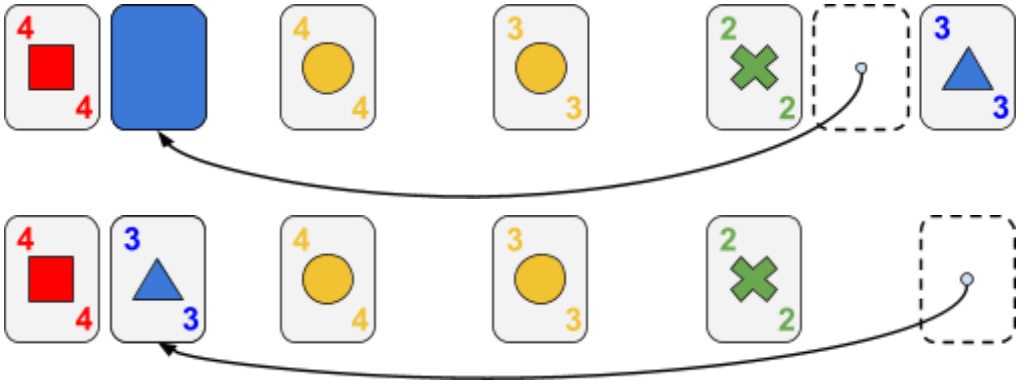


At this point, the only card remaining in the hand is Y4O. It cannot be placed on Pile #2, so the turn ends. The hand needs to be replenished with 6 cards from the draw pile so the player's hand is now: `Y4O B4X Y3O B4T G2X B B3T`



On the next turn, choose Pile #2 with the card B4T. Then play B4X, B, and B3T in that order. Since the Draw Pile is empty, the game ends.





There are 3 unplayable cards left in the hand (Y4O, Y3O, and G2X) and the draw pile is empty. The top cards in the piles are R4S on Pile #1 and B3T on Pile #2. The output is the string: `3 R4S B3T`.

INPUT: There will be three strings: the 2 face-up cards; the 7 cards in the player's hand; and the cards in the draw pile. There will be no more than 50 cards in the draw pile. All cards are separated by a single space.

OUTPUT: Output a string containing the number of cards that are left in the player's hand, the top card on Pile #1 and the top card on Pile #2, each separated by a single space.

SAMPLE INPUT:

```
1. R4S B3O
   R3X T B1T R1X B3X B1X Y4O
   B4X Y3O B4T G2X B B3T
2. G4T Y2S
   B4O G4X B1O G4O R4O R1O 1
   Y2T G3T Y4O G2T Y2O
3. R4T R3T
   G2S R4X B3O Y3X G3S Y3S Y4X
   B3T Y1T Y2T Y2O G2O B1T Y2S B2X B4X B2S Y1S R1O B1X B4S Y4S
4. G2X Y1X
   Y2O Y4T G2T B2O B2T Y3O B3X
   B1T B3O B1X G2S B1O G1X Y3S R3S Y1S B3S
5. G4O B1T
   4 B1O R4T B4T B3T B4X G1O
   G2S Y3T S R4S Y4S R Y
6. G2S B1T
   Y1X Y3O G4T 1 G4O B4T Y2X
   R4T S B1S B1O R2T B3T O G1O G3O R1O R1X
```

SAMPLE OUTPUT:

```
1. 3 R4S B3T
2. 1 Y2O G3T
3. 1 Y4S B3O
4. 6 B1O G1X
5. 3 G2S G1O
6. 7 B1O Y2X
```

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TEST DATA

TEST INPUT:

```
1. G1T B2S
   Y1T Y3T B3O B3S R3O G3S G3T
   Y1O B4T B1T Y4X B3T Y1X B1S B1X R1O
2. Y4T R2S
   B2S G2T R2X R1S Y2T B1S B2T
   G4T R3S B2X Y2S B1X G3T B1T Y3T
3. Y1S G3X
   R3X S B2X R2S B3X B B4O
   G4O 3 R1S R1T R3S G2X G B1T G3S B3T Y3S Y3O R1O R3O G1X R
   G4T Y1O
4. B1X Y3T
   R4X R1X B1O R3T R3X G1X R3S
   3 R4T Y4T G2S G4T Y3O Y3X G2O G G4S R1T B4T G1O S B1T R1S
   B2X
5. T R4X
   B Y1T B2X B1X Y2O R4O R2X
   R4T B4T R3T G1T G1X 1 R3O B3X Y Y1X Y3O Y4O G1O G4X
6. B4O Y
   B3X B2X B2O Y3O G3O Y1X Y1O
   Y3T G Y4T R1T 1 Y3X R3S G3S Y1T R4O Y2T 3 R R2T B1X R2X 2
   R1O B3S B1S R1X R1S Y1S G1O G4S Y2O G1S B4X Y2S R4S G4O X T
   B4S R2O Y4S B3O R3O S R3X 4 B2S O Y4O B B4T G2T G3T
```

TEST OUTPUT:

```
1. 1 Y4X B3T
2. 6 R2X R3S
3. 2 R G1X
4. 3 Y3X G2O
5. 4 R3T B3X
6. 0 R2O G3T
```

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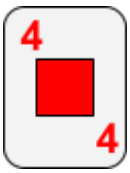
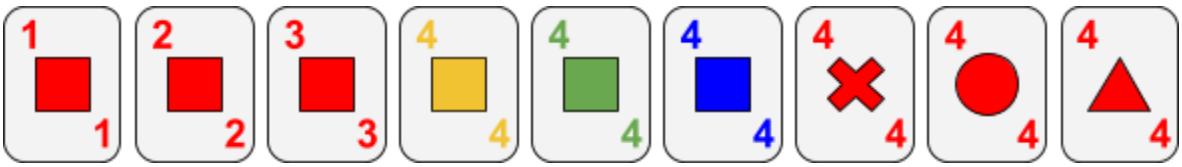
PROBLEM STATEMENT:

The card game *Duo* uses cards that have 3 characteristics: a color, number, and symbol. The game starts with two face-up cards and 7 cards in the player's hand. On a single turn, the player can choose one of the two face-up cards and *match* as many cards from their hand as possible on that pile. A card can be placed on top of another card if exactly 2 of the characteristics match.



The ACSL cards use four colors (R=red, Y=yellow, G=green, B=blue), four numbers (1, 2, 3, 4), and four symbols (X=cross, O=circle, S=square, T=triangle). A card is represented by a 3-character string showing the color, number, and symbol, in that order. For example, the card R4S, shown on the left, is red with the number 4 and contains a square. There are 9 cards that could possibly be placed on top of this card: a red square with the number 1, 2, or 3; a yellow, green, or blue square with the number 4; a red cross, circle, or triangle with the number 4.



The deck of cards may also have "wild" cards that have just 1 of these 3 characteristics and would be represented as a string such as a color (e.g. G) or a number (e.g. 3), or a shape (e.g. S). These cards can be matched if that one characteristic matches the other card's same characteristic, regardless of what the other two characteristics are.

Your program will play multiple turns of Uno Duo for a single player. The turn begins with two face-up cards, 7 cards in the hand (think of them as being held from left to right), and a draw pile.

1. Find the leftmost card in your hand that matches one of the piles. Place the matched card on that face-up card, creating a pile. If the card matches both face-up cards, choose the first card.
2. Try to match as many cards as you can in your hand on that pile. Each time, look at your cards starting at the leftmost.
3. If a Wild Card is at the top of the pile, play the first card that matches it.
4. Your turn is over when there are no more cards in your hand or you cannot play any of them.

When your turn is over, add as many cards as possible from the draw pile so that you have no more than 7 cards in your hand. Continue with another turn until you've played all the cards in your hand and the draw pile is empty, or no cards can be matched. Output the number of cards in your hand, followed by the top card in each pile. Separate each by a single space.

EXAMPLE:

Input:

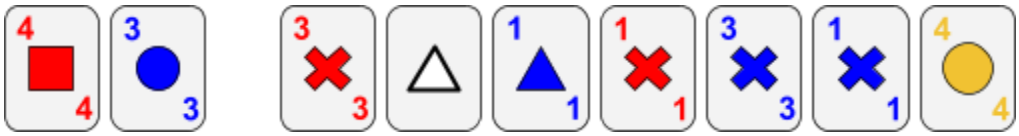
```
R4S B3O
R3X T B1T R1X B3X B1X Y4O
B4X Y3O B4T G2X B B3T
```

Output:

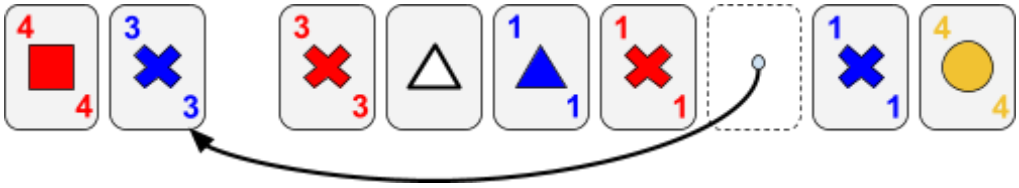
```
3 R4S B3T
```

Explanation:

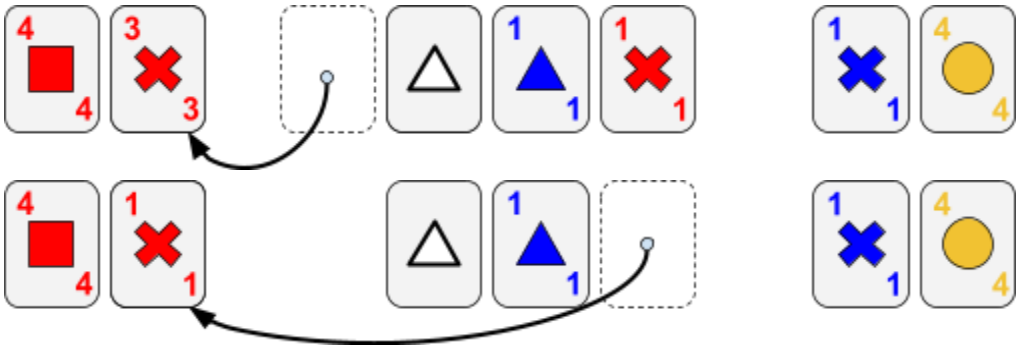
Initially, the two face-up cards and the 7 cards in the hand are as follows:

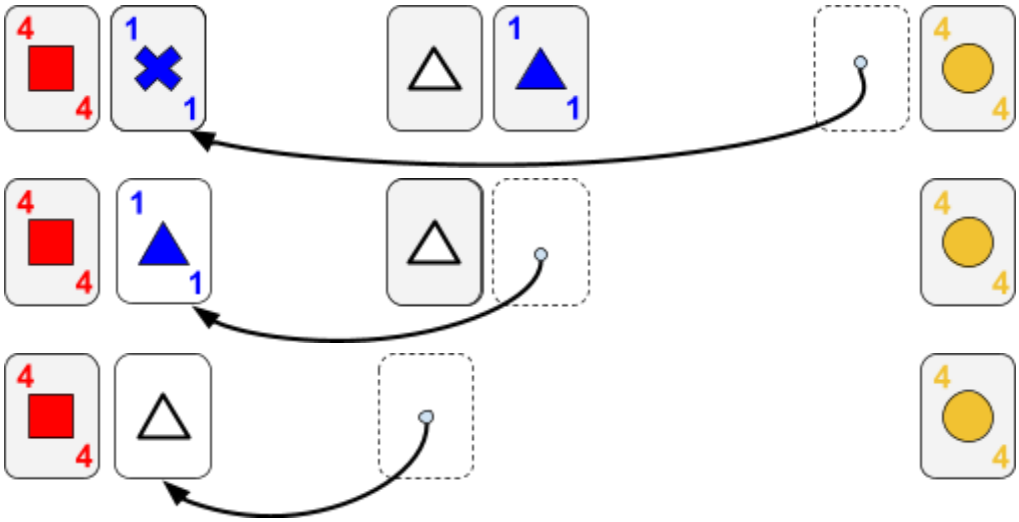


The first card in the hand that can be played is B3X (matching the color and the number, but not the shape) on the 2nd face-up card, thus making a pile. All future matches in this turn must go on the 2nd pile.

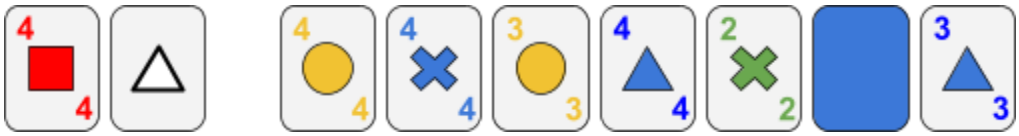


The next moves are as follows:

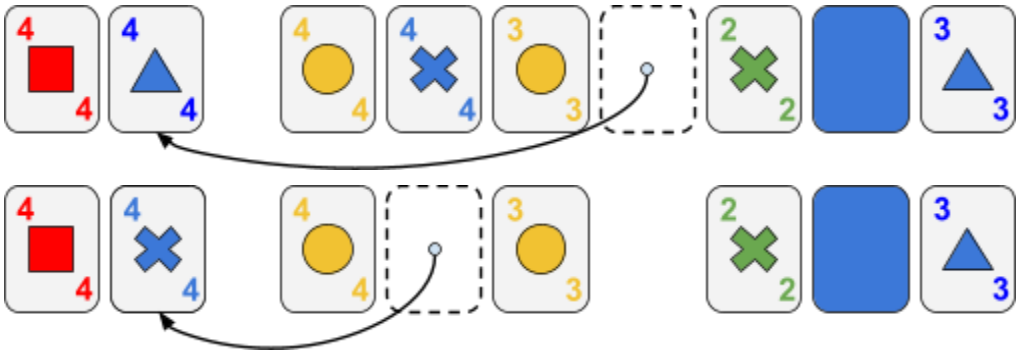




At this point, the only card remaining in the hand is Y4O. It cannot be placed on Pile #2, so the turn ends. At this point, the hand needs to be replenished with 6 cards from the draw pile so the player's hand is now: `Y4O B4X Y3O B4T G2X B B3T`



On the next turn, choose Pile #2 with the card B4T. Then play B4X, B, and B3T in that order. Since the draw pile is empty, the game ends.



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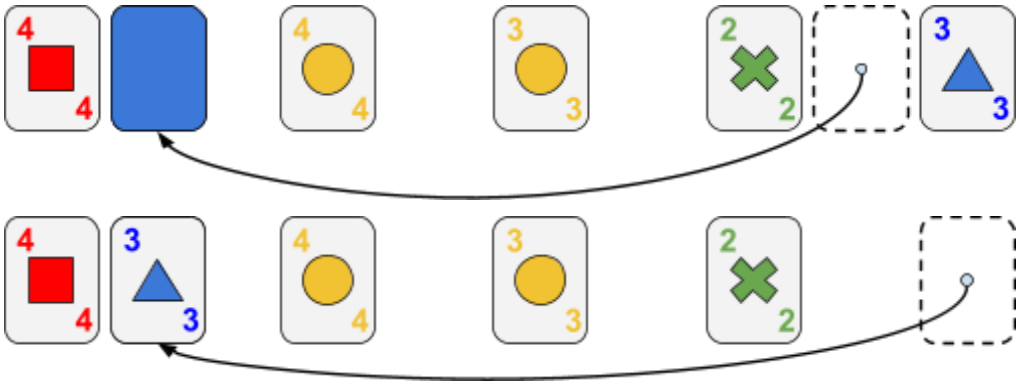

There are 3 unplayable cards left in the hand (Y4O, Y3O, and G2X) and the draw pile is empty. The top cards in the piles are R4S on Pile 1 and B3T on Pile 2. The output is the string: `3 R4S B3T`.

TASK:

Complete the function **printResult**

- The function has 3 parameters, all strings: the first, *piles*, representing the 2 face-up cards, the second, *hand*, representing the 7 cards in the player's hand, and the third, *drawPile*, representing the cards in the draw pile. All cards are separated by a single space.
- The function returns a string of the three values representing the number of cards left in the player's hand, the top card on Pile #1, and the top card on Pile #2, each separated by a single space.

You may create additional functions that are called from **printResult** if needed in solving the problem.

CONSTRAINTS:

Each card is represented by either a 3-character or 1-character string. There will be 2 face-up cards, 7 cards in the player's hand, and no more than 50 cards in the draw pile.

DATA PROVIDED:

There are 6 sets of Sample Data for debugging and 6 sets of Test Data for scoring. The test cases will vary in difficulty. You should create sample data of your own to fully test your program.

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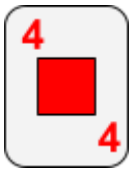
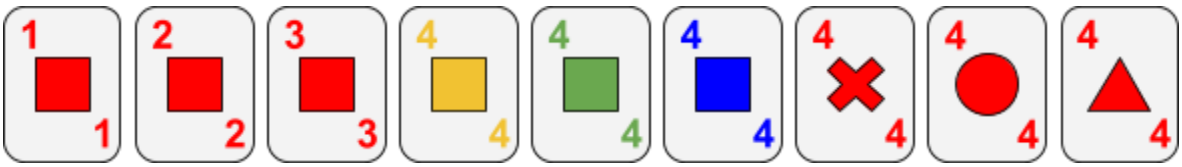
问题：*Duo* 卡牌游戏中，每张卡牌有 3 个特征：颜色、数字和符号。游戏开始时，有两张卡牌正面朝上，且玩家手中持有 7 张卡牌。在一轮游戏中，玩家可以从两张正面朝上的卡牌中选择一张，并尽可能多地将手中的卡牌与该牌堆上的卡牌进行*匹配*。如果两张卡牌恰好有两个特征相匹配，就可以将一张卡牌放在另一张卡牌的上面。



ACSL 版卡牌使用四种颜色（R = 红色，Y = 黄色，G = 绿色，B = 蓝色）、四个数字（1、2、3、4）和四种符号（X = 叉号，O = 圆形，S = 正方形，T = 三角形）。每张卡牌用一个包含 3 个字符的字符串表示，并按照颜色、数字、符号的顺序依次排列。例如，左侧所示的卡牌 R4S 是一张颜色为红色，数字为 4，符号为正方形的卡牌。有 9 张卡牌可能会被放在这张卡牌的上面：颜色为红色，数字为 1、2 或 3，符号为正方形的卡牌；颜色为黄色、绿色或蓝色，数字为 4 ，符号为正方形的卡牌；颜色为红色，数字为 4 ，符号为叉号、圆形或三角形的卡牌。



牌堆中可能还有 “万能” 卡牌，这种卡牌只具有 3 个特征中的其中 1 个特征，并以颜色（例如 G）、数字（例如 3）或符号（例如 S）这样的字符串表示。如果万能卡牌的这一特征与另一张卡牌的同一特征匹配，则无论其他两个特征是什么，这些卡牌都可以进行匹配。

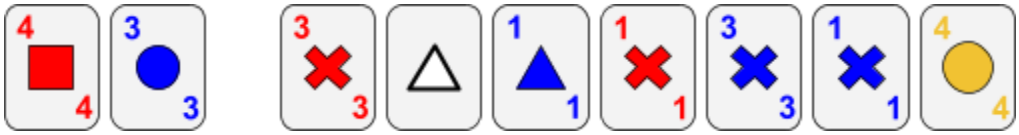
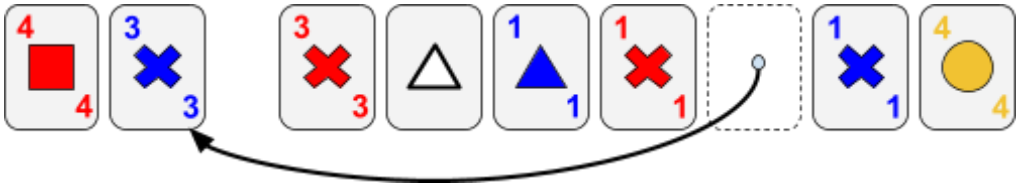
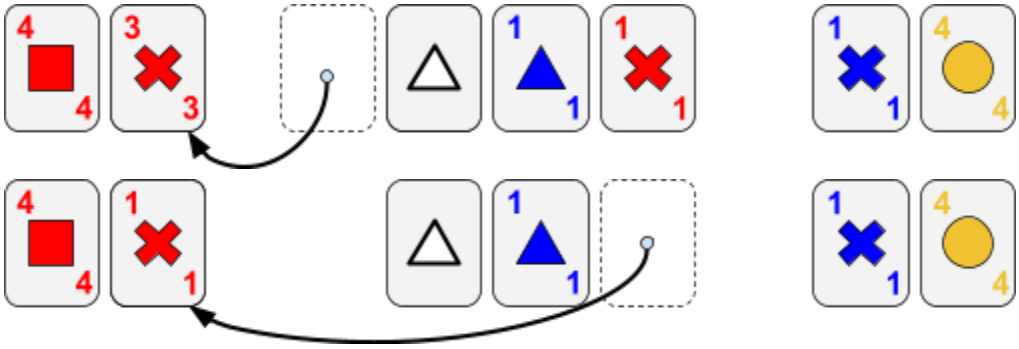
你的程序将模拟单个玩家在多轮 Uno Duo 游戏中的操作。每一轮游戏开始时，有两张卡牌正面朝上，玩家手中持有 7 张卡牌（可将卡牌顺序视为从左至右排列），以及一个抽牌堆。

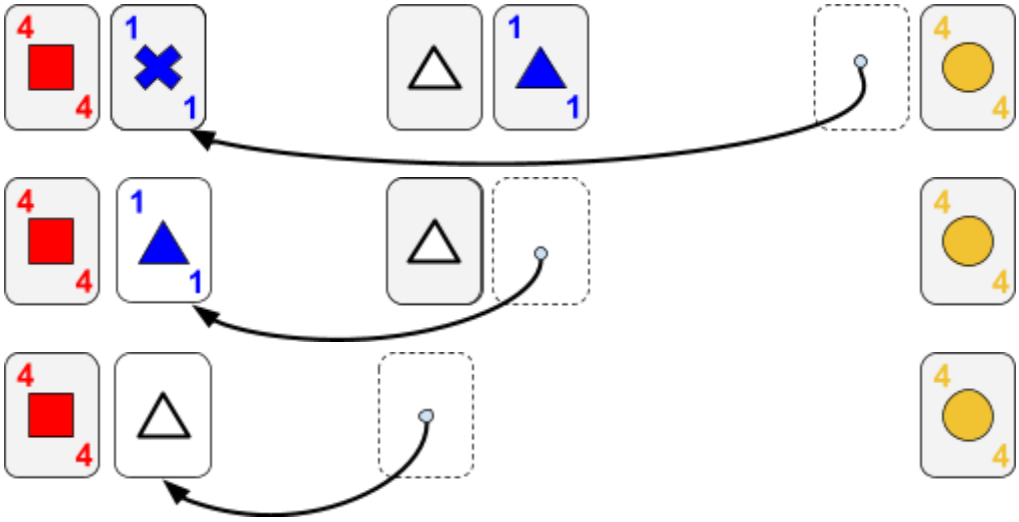
1. 从手牌中找到最左侧一张与两张正面朝上卡牌中的其中一张相匹配的卡牌。将匹配的卡牌放置在那张正面朝上的卡牌上面，形成一个牌堆。如果该卡牌与两张正面朝上的卡牌都匹配，则放在第一张卡牌上面。
2. 尝试将手中尽可能多的卡牌与该牌堆上的卡牌进行匹配。每次都从最左侧的卡牌开始查看。
3. 如果牌堆最上面是一张万能卡牌，则将与之匹配的第一张卡牌放在它上面。
4. 当手中没有卡牌或卡牌无法再继续匹配时，本轮游戏结束。

一轮游戏结束后，从抽牌堆中抽取尽可能多的纸牌，但手中持有的卡牌数不能超过 7 张。然后进行下一轮游戏，直到手中和抽牌堆中没有卡牌，或者没有卡牌可以再继续匹配。输出手中剩余卡牌的数量，后接每个牌堆最上面的卡牌。每张卡牌之间用一个空格隔开。

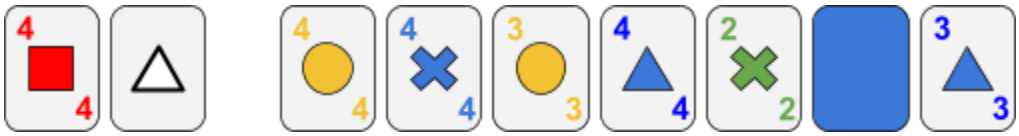
样例：

输入	`R4S B3O`
	`R3X T B1T R1X B3X B1X Y4O`
	`B4X Y3O B4T G2X B B3T`

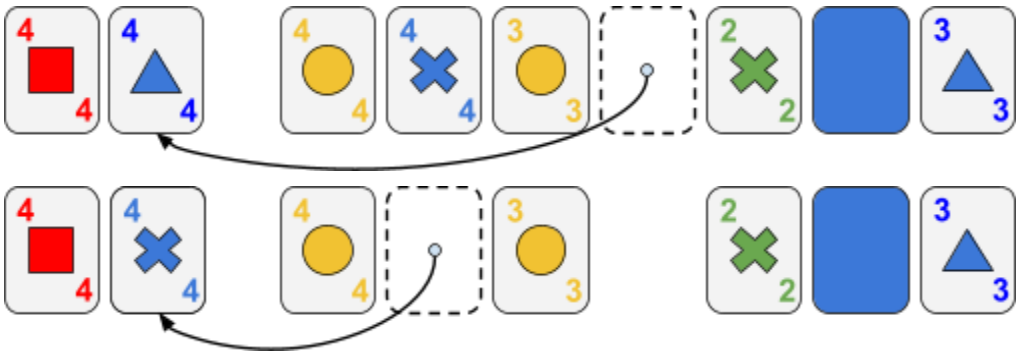
输出	`3 R4S B3T`
解释说明	游戏开始时，两张正面朝上的卡牌和玩家手中持有的 7 张卡牌如下所示：  手中可以匹配的第一张卡牌是 B3X（颜色和数字匹配，但符号不匹配），将其放在第二个正面朝上的卡牌上面，从而形成一个牌堆。在这一轮游戏中，后续所有匹配的卡牌都必须放在第二个牌堆上面。  接下来的卡牌匹配情况如下所示： 

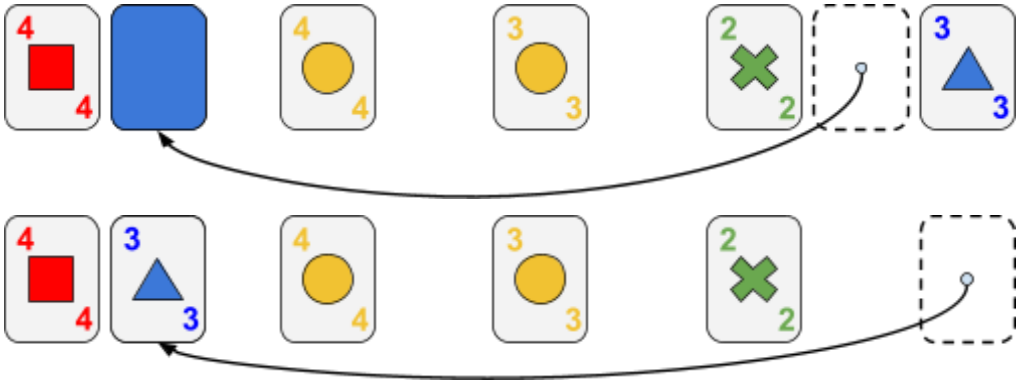


此时，玩家手中唯一剩下的卡牌是 Y4O。该卡牌无法放置在牌堆 2 上面，因此本轮游戏结束。玩家需要从抽牌堆中抽取 6 张卡牌补充手中的卡牌，此时玩家手中持有的卡牌为：`Y4O B4X Y3O B4T G2X B B3T`



在下一轮游戏中，选择牌堆 2 进行匹配，其顶部卡牌为 B4T。然后依次放置 B4X、B 和 B3T。由于抽牌堆中再无卡牌，因此游戏结束。



	 手中剩下 3 张无法再继续匹配的卡牌 (Y4O、Y3O、G2X)，且抽牌堆中再无卡牌。牌堆 1 最上面的卡牌是 R4S，牌堆 2 最上面的卡牌是 B3T。输出的字符串为：`3 R4S B3T.`

输入：将有三个字符串，分别是：两张正面朝上的卡牌；玩家手中持有的 7 张卡牌；抽牌堆中的卡牌。抽牌堆中最多有 50 张卡牌。每张卡牌之间用一个空格隔开。

输出：输出一个字符串，包含玩家手中剩余卡牌的数量、牌堆 1 最上面的卡牌以及牌堆 2 最上面的卡牌，每个卡牌之间用一个空格隔开。

样例输入：

```
1. R4S B3O
   R3X T B1T R1X B3X B1X Y4O
   B4X Y3O B4T G2X B B3T
2. G4T Y2S
   B4O G4X B1O G4O R4O R1O 1
   Y2T G3T Y4O G2T Y2O
3. R4T R3T
   G2S R4X B3O Y3X G3S Y3S Y4X
   B3T Y1T Y2T Y2O G2O B1T Y2S B2X B4X B2S Y1S R1O B1X B4S Y4S
4. G2X Y1X
   Y2O Y4T G2T B2O B2T Y3O B3X
   B1T B3O B1X G2S B1O G1X Y3S R3S Y1S B3S
5. G4O B1T
   4 B1O R4T B4T B3T B4X G1O
   G2S Y3T S R4S Y4S R Y
6. G2S B1T
   Y1X Y3O G4T 1 G4O B4T Y2X
   R4T S B1S B1O R2T B3T O G1O G3O R1O R1X
```

样例输出：

```
1. 3 R4S B3T
2. 1 Y2O G3T
3. 1 Y4S B3O
4. 6 B1O G1X
5. 3 G2S G1O
6. 7 B1O Y2X
```

American Computer Science League

2024-2025 ● Contest 2: Uno Duo ● Senior Division

测试数据

测试输入：

```
1. G1T B2S
   Y1T Y3T B3O B3S R3O G3S G3T
   Y1O B4T B1T Y4X B3T Y1X B1S B1X R1O
2. Y4T R2S
   B2S G2T R2X R1S Y2T B1S B2T
   G4T R3S B2X Y2S B1X G3T B1T Y3T
3. Y1S G3X
   R3X S B2X R2S B3X B B4O
   G4O 3 R1S R1T R3S G2X G B1T G3S B3T Y3S Y3O R1O R3O G1X R
   G4T Y1O
4. B1X Y3T
   R4X R1X B1O R3T R3X G1X R3S
   3 R4T Y4T G2S G4T Y3O Y3X G2O G G4S R1T B4T G1O S B1T R1S
   B2X
5. T R4X
   B Y1T B2X B1X Y2O R4O R2X
   R4T B4T R3T G1T G1X 1 R3O B3X Y Y1X Y3O Y4O G1O G4X
6. B4O Y
   B3X B2X B2O Y3O G3O Y1X Y1O
   Y3T G Y4T R1T 1 Y3X R3S G3S Y1T R4O Y2T 3 R R2T B1X R2X 2
   R1O B3S B1S R1X R1S Y1S G1O G4S Y2O G1S B4X Y2S R4S G4O X T
   B4S R2O Y4S B3O R3O S R3X 4 B2S O Y4O B B4T G2T G3T
```

测试输出：

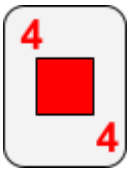
```
1. 1 Y4X B3T
2. 6 R2X R3S
3. 2 R G1X
4. 3 Y3X G2O
5. 4 R3T B3X
6. 0 R2O G3T
```

American Computer Science League

2024-2025 ● Contest 2: Uno Duo ● Senior Division

问题陈述：

Duo 卡牌游戏中，每张卡牌有 3 个特征：颜色、数字和符号。游戏开始时，有两张卡牌正面朝上，且玩家手中持有 7 张卡牌。在一轮游戏中，玩家可以从两张正面朝上的卡牌中选择一张，并尽可能多地将手中的卡牌与该牌堆上的卡牌进行*匹配*。如果两张卡牌恰好有两个特征相匹配，就可以将一张卡牌放在另一张卡牌的上面。



ACSL 版卡牌使用四种颜色（R = 红色，Y = 黄色，G = 绿色，B = 蓝色）、四个数字（1、2、3、4）和四种符号（X = 叉号，O = 圆形，S = 正方形，T = 三角形）。每张卡牌用一个包含 3 个字符的字符串表示，并按照颜色、数字、符号的顺序依次排列。例如，左侧所示的卡牌 R4S 是一张颜色为红色，数字为 4，符号为正方形的卡牌。有 9 张卡牌可能可以放在这张卡牌的上面：颜色为红色，数字为 1、2 或 3，符号为正方形的卡牌；颜色为黄色、绿色或蓝色，数字为 4 ，符号为正方形的卡牌；颜色为红色，数字为 4 ，符号为叉号、圆形或三角形的卡牌。

牌堆中可能还有 “万能” 卡牌，这种卡牌只具有 3 个特征中的其中 1 个特征，并以颜色（例如 G）、数字（例如 3）或符号（例如 S）这样的字符串表示。如果万能卡牌的这一特征与另一张卡牌的同一特征匹配，则无论其他两个特征是什么，这些卡牌都可以进行匹配。

你的程序将模拟单个玩家在多轮 Uno Duo 游戏中的操作。每一轮游戏开始时，有两张卡牌正面朝上，玩家手中持有 7 张卡牌（可将卡牌顺序视为从左至右排列），以及一个抽牌堆。

1. 从手牌中找到最左侧一张与两张正面朝上卡牌中的其中一张相匹配的卡牌。将匹配的卡牌放置在那张正面朝上的卡牌上面，形成一个牌堆。如果该卡牌与两张正面朝上的卡牌都匹配，则放在第一张卡牌上面。
2. 尝试将手中尽可能多的卡牌与该牌堆上的卡牌进行匹配。每次都从最左侧的卡牌开始查看。
3. 如果牌堆最上面是一张万能卡牌，则将与之匹配的第一张卡牌放在它上面。
4. 当手中没有卡牌或卡牌无法再继续匹配时，本轮游戏结束。

一轮游戏结束后，从抽牌堆中抽取尽可能多的纸牌，但手中持有的卡牌数不能超过 7 张。然后进行下一轮游戏，直到手中和抽牌堆中没有卡牌，或者没有卡牌可以再继续匹配。输出手中剩余卡牌的数量，后接每个牌堆最上面的卡牌。每张卡牌之间用一个空格隔开。

样例：

输入：

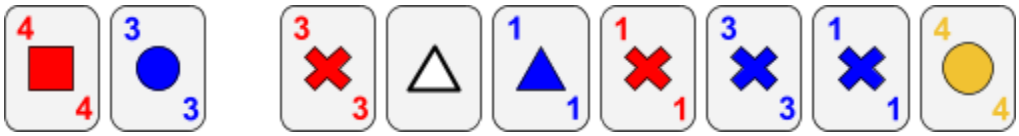
```
R4S B3O
R3X T B1T R1X B3X B1X Y4O
B4X Y3O B4T G2X B B3T
```

输出：

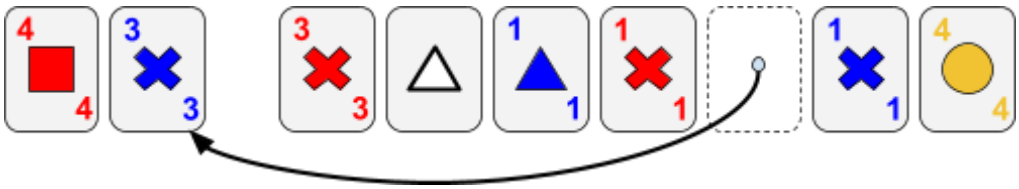
```
3 R4S B3T
```

解释说明：

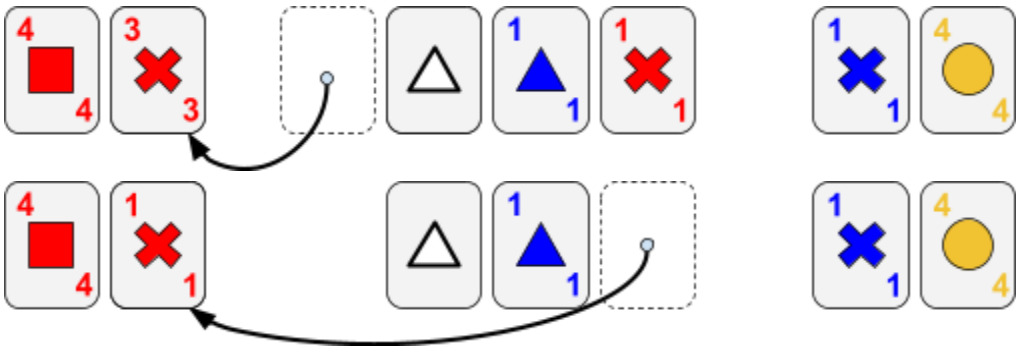
游戏开始时，两张正面朝上的卡牌和玩家手中持有的 7 张卡牌如下所示：

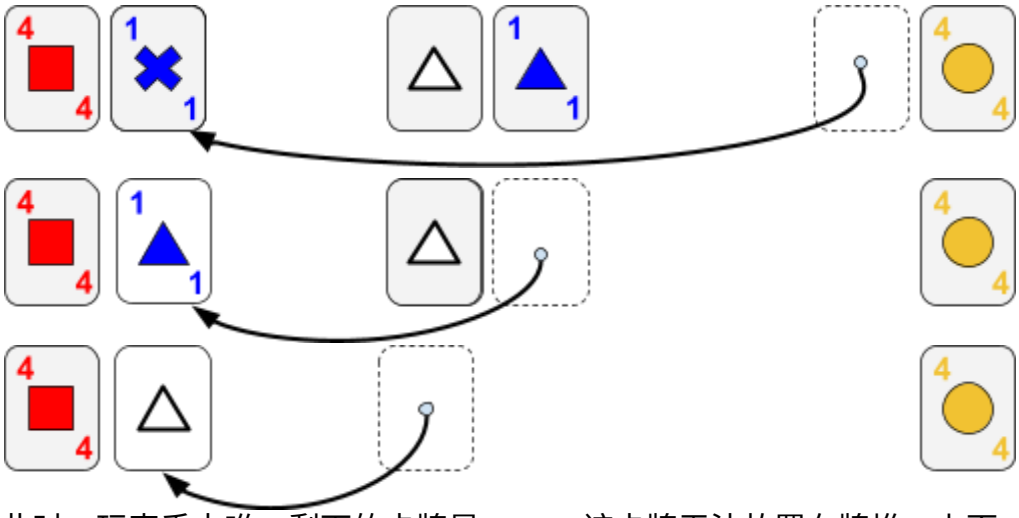


手中可以匹配的第一张卡牌是 B3X（颜色和数字匹配，但符号不匹配），将其放在第二个正面朝上的卡牌上面，从而形成一个牌堆。在这一轮游戏中，后续所有匹配的卡牌都必须放在第二个牌堆上面。



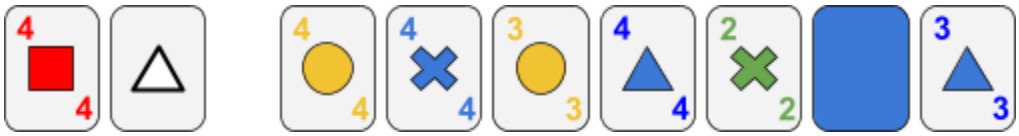
接下来的卡牌匹配情况如下所示：



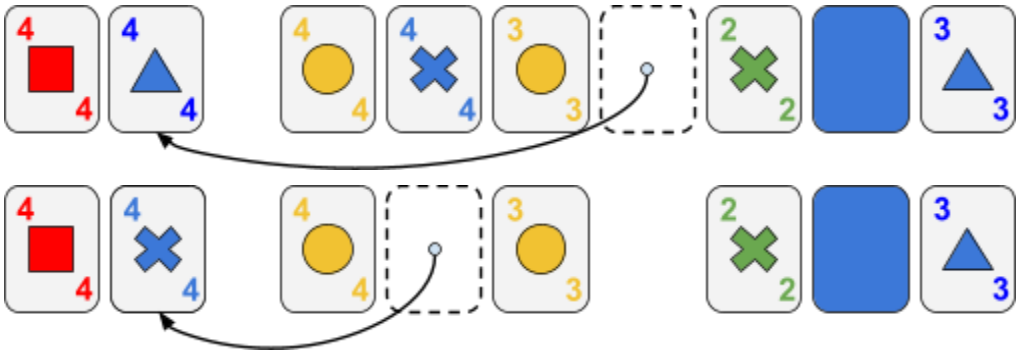


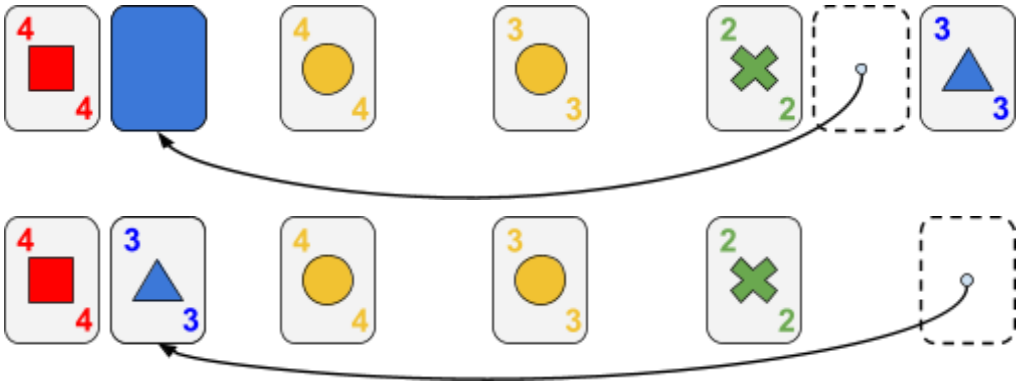
此时，玩家手中唯一剩下的卡牌是 Y4O。该卡牌无法放置在牌堆 2 上面，因此本轮游戏结束。玩家需要从抽牌堆中抽取 6 张卡牌补充手中的卡牌，此时玩家手中持有的卡牌为：

`Y4O B4X Y3O B4T G2X B B3T`



在下一轮游戏中，选择牌堆 2 进行匹配，其顶部卡牌为 B4T。然后依次放置 B4X、B 和 B3T。由于抽牌堆中再无卡牌，因此游戏结束。





手中剩下 3 张无法再继续匹配的卡牌 (Y4O、Y3O、G2X)，且抽牌堆中再无卡牌。牌堆 1 最上面的卡牌是 R4S，牌堆 2 最上面的卡牌是 B3T。输出的字符串为：`3 R4S B3T`.

任务：

完成函数 **printResult**

- 该函数有 3 个参数，均为字符串：第一个参数 *piles* 表示两张正面朝上的卡牌，第二个参数 *hand* 表示玩家手中的 7 张卡牌，第三个参数 *drawPile* 表示抽牌堆中的卡牌。所有卡牌之间用一个空格隔开。
- 该函数返回一个由三个值组成的字符串，这三个值分别表示玩家手中剩余的卡牌数量、牌堆 1 最上面的卡牌以及牌堆 2 最上面的卡牌，且每个卡牌之间用一个空格隔开。

如有需要，你可以构建额外的函数供 **printResult** 调用来解答此问题。

约束条件：

每张卡牌由一个包含 3 个字符或 1 个字符的字符串表示。有两张正面朝上的卡牌和玩家手中持有的 7 张卡牌，且抽牌堆中最多有 50 张卡牌。

提供的数据：

有 6 组用于调试的样例数据和 6 组用于评分的测试数据。测试用例的难度各不相同。
你应该创建自己的样例数据来充分测试你的程序。